

# Ekansh Aryan Sinha

Bangalore, India | +91 6201524943 | sekansharyan2003@gmail.com | linkedin.com/in/ekansh-aryan-sinha

## SOFTWARE DEVELOPMENT PROFILE

Final-year B.E. Electronics and Communication Engineering and MSc Economics student at BITS Pilani with internship experience in backend microservices, REST APIs, Python automation, quantitative analytics, debugging, and production-style software development. Strong CS foundation in DSA, OOP, OS, DBMS, and Computer Networks; solved 500+ LeetCode problems.

## EDUCATION

**Birla Institute of Technology and Science Pilani, Hyderabad Campus** Hyderabad, India  
*B.E. Electronics and Communication Engineering & MSc Economics; CGPA: 8.03/10* *Aug 2021 – May 2026*  
*Relevant Coursework:* Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks, Internet of Things

## EXPERIENCE

**Nykaa** Bangalore, India  
*Software Development Engineer Intern* *Jul 2025 – Dec 2025*

- Built a FastAPI-based mock API microservice, reducing test setup effort by 40% and improving developer productivity for backend testing workflows.
- Developed data synchronization workflows for hierarchical datasets across backend modules, improving consistency, maintainability, and reliability of test data flows.
- Added structured logging, metrics, validation, and error-handling support to improve observability, debugging efficiency, and API workflow reliability.

**MerQube** Bangalore, India  
*Quantitative Analyst Intern* *Jan 2026 – Present*

- Designed and implemented an options-based index strategy in Python with the VIX Index as the underlying, targeting volatility-driven return opportunities through systematic index construction.
- Owned end-to-end index development, including strategy logic, portfolio generation, market-data-driven calculations, validation, debugging, and production-ready implementation.
- Optimized index calculation workflows by adding intermediate-result caching, portfolio persistence, and reusable configuration-driven components to improve runtime and maintainability.
- Collaborated directly with clients to investigate reported issues, identify root causes, clarify requirements, and coordinate corrections across internal implementation and client-side inputs.

## TECHNICAL SKILLS

**Programming Languages:** Python, C++, JavaScript, SQL

**Backend Development:** FastAPI, REST APIs, Microservices, Backend Testing

**Databases:** SQL, MongoDB, Relational Schema Design

**Tools & Platforms:** Git, Docker, Postman, Jupyter Notebook, Microsoft Excel

**CS Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Multithreading, TCP/IP, HTTP

**Quantitative & Analytics:** Python Automation, Market Data Analysis, Derivatives Analytics, Index Calculation Workflows, Data Validation, EC2, S3

## PROJECTS

**Online Food Delivery System** | *Database Design, SQL, NoSQL* Sep 2024 – Dec 2024

- Designed SQL schema with triggers, stored procedures, and ACID transactions; implemented a NoSQL model for read-heavy workloads, improving query latency by 50% and reducing storage by 75%.

**Time Table Builder** | *Java, Object-Oriented Programming, Multithreading* Jan 2025 – Apr 2025

- Developed a Java scheduling engine with modular components for rule-based validation, slot allocation, CSV ingestion, and conflict-free timetable generation; achieved 95% scheduling accuracy.

**Port Management System** | *C, Operating Systems, Multithreading, IPC* Feb 2025 – Apr 2025

- Engineered a concurrent port management system using message queues and shared memory to coordinate scheduling and validation, processing 567 ships in under 3 minutes while preventing race conditions.

**Chat Management and Moderation System** | *C, POSIX IPC, Semaphores* Jan 2025 – Mar 2025

- Built a real-time chat system with user authorization, group-level moderation, message ordering guarantees, semaphore-based synchronization, and automated violation handling.

## ACHIEVEMENTS & CERTIFICATIONS

- Solved 500+ LeetCode problems across data structures, algorithms, dynamic programming, graphs, trees, arrays, and strings.
- Achieved global rank 690 out of 34,000+ participants in a LeetCode Weekly Contest.
- Completed Machine Learning Specialization by Andrew Ng, Google Cloud Fundamentals: Core Infrastructure, and Google AI Essentials.